🇮🇳 Yojana Sahayak योजना सहायक

"Ask in your language. Hear only what's verified."

✓ ZERO-HALLUCINATION VERIFIED

VOICE-FIRST GOVTECH INNOVATION

⚠ THE WELFARE DELIVERY GAP

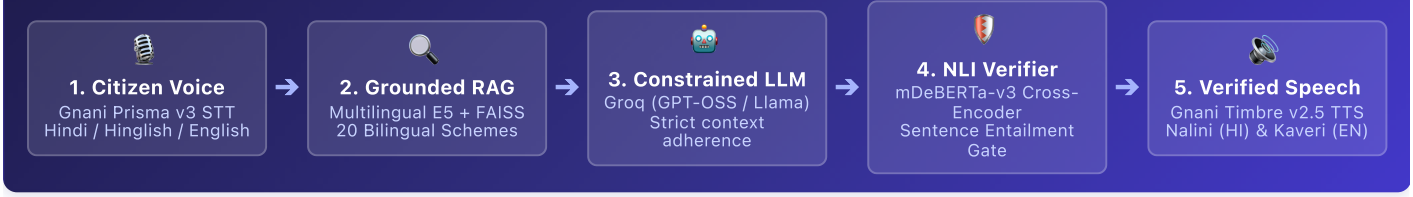
- **300M+ eligible citizens** miss out on benefits (PM-KISAN, Ujjwala, PMAY) due to bureaucratic PDFs and language/literacy barriers.
- **Commercial LLM Hallucinations:** Standard generative models invent rules, deadlines, and rupee amounts, leading to rejected applications.
- **Intermediary Fraud:** Citizens face exploitation by unauthorized middlemen charging fees for free public services.

💡 THE SOLUTION: YOJANA SAHAYAK

- **Voice-First Inclusion:** Natural spoken Hindi/English voice input and playback for citizens regardless of digital literacy.
- **NLI Verification Gate:** Every generated sentence is checked against retrieved official passages using cross-encoder inference.
- **Verified-Only Audio:** Only sentences strictly verified as *Supported* are spoken aloud; unverified statements are held back.

END-TO-END ARCHITECTURAL PIPELINE

Closed-Loop Grounded RAG + Multilingual Voice Verification



CORE TECHNICAL INNOVATIONS & DIFFERENTIATORS

Chunk-Aware Verification Gate

Unlike naive RAG, our verifier scopes entailment and contradiction by retrieved scheme chunk. It eliminates false cross-section contradictions (e.g. Overview vs. Exclusion lists) while maintaining 100% precision against hallucinations.

Strict Verified-Only Audio Contract

If any sentence fails verification, voice playback of unverified claims is suppressed. The system speaks a transparent fallback disclaimer pointing to official portals, ensuring citizens are never exposed to audio misinformation.

Transparent Side-by-Side Audit Panel

Every answer is paired with an Evidence Panel displaying the exact official government source chunks, official portal links (e.g. `pmkisan.gov.in`), overall groundedness score, and sentence-by-sentence checks (Supported / Contradicted).

Native Indian Language Engineering

Built with domain keyword boosting for 200+ Indian welfare terms (eKYC, KCC, DBT, Samman Nidhi). Includes spoken number and currency normalization (converting ₹6,000 to "छह हज़ार रुपये") for crystal-clear rural comprehension.

DEPLOYMENT SCOPE & SCALABILITY

20	100%	40/40	500K+
Bilingual Schemes Indexed	Grounded Speech Guarantee	Benchmark Split Validated	Target CSC Touchpoints

TECHNOLOGY STACK

Gnani Prisma v3 STT · Gnani Timbre v2.5 TTS · intfloat/multilingual-e5-base · FAISS IndexFlatIP · mDeBERTa-v3 NLI · Groq / Ollama Llama 3.1 · Streamlit 1.67 · PyTorch / HuggingFace